Quadratic Programming

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Overview

Formulation

- A quadratic program (QP) is an optimization problem of the form

$$min_{x \in \mathbb{R}^n} \frac{1}{2} x^T G x + c^T x \quad \text{subject to} \quad \begin{aligned} a_i^T x - b_i &= 0, \; i \in \mathcal{E} \\ a_i^T x - b_i &\geq 0 \; i \in \mathcal{I} \end{aligned}$$

- G: symmetric $n \times n$ matrix
- $c \in \mathbb{R}^n$: a vector
- $\mathcal{E}$: index for **equality constraints**
- $\mathcal{I}$: index for **inequality constraints**
- When $G = 0$, it reduces to a linear program

QP with only equality constraints

Consider a QP with only equality constraints as

$$min_{x\in\mathbb{R}^n} \frac{1}{2}x^T Gx + c^T x \quad \text{subject to } \; Ax = b \tag{1}$$

Comparing it with a general constrained optimization problem, we have

- Objective function: $f(x) = \frac{1}{2}x^T Gx + c^T x$
- Objective gradient: $\nabla f(x) = Gx + c$
- Lagrange multiplier: $\lambda \in \mathbb{R}^m$
- Constraint gradient: $c_i(x) = a_i^T x - b_i$
- Constraint gradient: $\nabla c_i(x) = a_i$
- Lagrange function: $L = f - \sum_i \lambda_i c_i$

KKT Conditions

The KKT conditions for the QP with m equality constraints is

- Gradient condition: $Gx + c = A^T\lambda$
- Feasibility: $Ax = b$
- NOTE: Since there are no inequality constraints, there are no conditions that the Lagrange multipliers be non-negative or complementary slackness
- The above equation can be reformulated as

$$\begin{bmatrix} G & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix} \tag{2}$$

- Equation 2 is a linear algebra problem

QP: Inequality constraints

Consider now the general QP in (1)

- Let (x^*, λ^*) be the optimal solution for the constrained problem
- Now, one needs to deal with complementary slackness - some of the inequality constraints are active at the optimum $(a_i^T x^* = b_i)$ and only those Lagrangian multipliers can be non-zero
- Let $\mathcal{A}(x^*)$ be those indices in $\mathcal{E}$ and those in $\mathcal{I}$ such that the inequality constraint is active at the optimum (x^*, λ^*)- i.e.

$$\mathcal{A}(x^*) := \{i \in \mathcal{E} \cup \mathcal{I} \mid a_i^T x^* = b_i\} \tag{3}$$

  Then, $\lambda_i^* \geq 0$ for $i \in \mathcal{A}(x^*) \cap \mathcal{I}$
- The Lagrange multipliers corresponding to the remaining inactive inequality constraints should vanish - i.e. $\lambda_i^* = 0$ for $i \notin \mathcal{A}(x^*)$

KKT Conditions

- If somehow we knew $\mathcal{A}(x^*)$ in advance, KKT conditions, after putting $\lambda_i^* = 0$ for $i \notin \mathcal{A}(x^*)$, become
  - Gradient condition:
$$Gx^* + c - \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* a_i = 0 \tag{4}$$
  - Equality feasiblity:
$$a_i^T x^* = b_i \quad \text{for } i \in \mathcal{A}(x^*) \tag{5}$$
  - Inequality feasiblity:
$$a_i^T x^* \geq b_i \quad \text{for } i \notin \mathcal{A}(x^*) \cap \mathcal{I} \tag{6}$$
  - Lagrange multiplier:
$$\lambda_i^* \geq 0 \quad \text{for } i \in \mathcal{A}(x^*) \cap \mathcal{I} \tag{7}$$

Note

- One sees that once $\mathcal{A}(x^*)$ is known, equations (4)-(5) amount to a linear algebra problem for (x^*, λ^*) and verification of the inequalities (6)-(10)
- Consider the following steps
  - Guess an $\mathcal{A}(x^*)$
  - Solve the equations (4)-(5) to compute (x^*, λ_i^*) where $i \in \mathcal{A}(x^*)$
  - Verify the inequalities (6)-(10)
- Suppose if by chance, the inequalities (6)-(10) are satisfied for the solved (x^*, λ_i^*), then all the KKT conditions are satisfied and hence the computed x^* is optimal!
- Note that there are $n + \mathcal{A}(x^*)$ unknowns (x, λ_i^*) and $n + \mathcal{A}(x^*)$ linear equations in (4)-(5) and hence (4)-(5) is a standard linear algebra problem where the number of unknowns match the number of equations

Active Set Method algorithm: Motivation and Intuition

- The active set method algorithms proceed by guessing an $\mathcal{A}(x^*)$ as above
  - Guess an $\mathcal{A}(x^*)$
  - Solve the equations (4)-(5) to compute (x^*, λ_i^*) where $i \in \mathcal{A}(x^*)$
  - Verify the inequalities (6)-(10)
- If (6)-(10) are all satisfied, STOP (x^* is the optimum)
- If not (some inequality constraint is violated), then one can guess another $\mathcal{A}(x^*)$ and proceed such that the objective function decreases for the new x^* computed from the new $\mathcal{A}(x^*)$ - intuition for **active set method** algorithms
- One proceeds till terminating condition is reached
- **NOTE:** This is similar to the simplex algorithm in LP where the basic feasible point set $\mathcal{B}$ is guessed and corrected for iteratively

Convex QP

- A QP is called a convex QP if the quadratic G in the convex function is positive semidefinite - this results in the objective function being convex
- For convex QPs, the KKT conditions above are **necessary and sufficient** for optima

Reformulating the constraints

- Consider the problems (4)-(5) given a set $i \in \mathcal{A}(x^*)$.

$$Gx^* + c - \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* a_i = 0$$
$$a_i^T x^* = b_i$$

- Then, the above equations solve the problem

$$min_x \left(\frac{1}{2} x^T G x + c^T x \right)$$
$$\text{subject to:} \quad a_i^T x = b_i \quad \text{for } i \in \mathcal{A}(x^*)$$

Reformulating the constraints

- Let $x = x_0 + p$ where x_0 is a guess point that is feasible - i.e. $a_i^T x_0 = b_i$ for $i \in \mathcal{A}(x^*)$ and $a_i^T x_0 \geq b_i$ for $i \notin \mathcal{A}(x^*)$. Then , the problem reduces to

$$min_p \left(\frac{1}{2} p^T G p + g_0^T p + \rho_0 \right) \tag{8}$$

$$\text{subject to:} \quad a_i^T p = 0 \quad \text{for } i \in \mathcal{A}(x^*) \tag{9}$$

where

$$g_0 = G x_0 + c$$

$$\rho_0 = \frac{1}{2} x_0^T G x_0 + c^T x_0$$

- If p_0 that solves (8)-(9) also satisfies all the inequality constraints in (6)-(10) as below, then $x^* = x_0 + p_0$ is optimal

$$a_i^T (x_0 + p_0) \geq b_i \quad \text{for } i \notin \mathcal{A}(x^*) \cap \mathcal{I}$$

$$\lambda_i^* \geq 0 \quad \text{for } i \in \mathcal{A}(x^*) \cap \mathcal{I}$$

The Algorithm - Step 1

Inputs to k^{th} iteration:
x_k - feasible starting value
$\mathcal{W}_k$ - starting guess for $\mathcal{A}(x^*)$
Compute:
$g_k = Gx_k + c$ and $\rho_k = \frac{1}{2}x_k^T Gx_k + c^T x_k$

- Solve for p_k - Equation (L)

$$min_{p_k}\left(\frac{1}{2}p_k^T Gp_k + g_k^T p_k + \rho_k\right)$$
$$\text{subject to:} \quad a_i^T p_k = 0 \quad \text{for } i \in \mathcal{W}_k$$

Case 1: $p_k = 0$

If solution p_k to (L) is 0, then compute λ_i for $i \in \mathcal{W}_k \cap \mathcal{I}$

- If $\lambda_i \geq 0$ for all $i \in \mathcal{W}_k \cap \mathcal{I}$,
  then set $x^* = x_k$ and STOP - **termination condition**
- Else if for some $j' \in \mathcal{W}_k \cap \mathcal{I}$, $\lambda_j' < 0$,
  then set $j := argmin_{(j' \in \mathcal{W}_k \cap \mathcal{I})} \lambda_{j'}$
  - **For next iteration:** Remove j from $\mathcal{W}_k$ and proceed
    $\mathcal{W}_{k+1} = \mathcal{W}_k / \{j\}$ and $x_{k+1} = x_k$

Case 2: $p_k \neq 0$

If solution p_k to (L) is non-zero, then

- $x_{k+1} = x_k + \alpha_k p_k$, find the lowest step length $\alpha_k \in [0, 1]$ that renders one constraint not in $\mathcal{W}_k$, active - i.e. Find

$$\alpha_k := \min \left(1, \min_{i \notin \mathcal{W}_k, a_i^T p_k < 0} \frac{b_i - a_i^T x_k}{a_i^T p_k} \right) \tag{10}$$

- Check if $\alpha_k < 1$
  - If $\alpha_k < 1$, then $\alpha_k = \min_{i \notin \mathcal{W}_k, a_i^T p_k < 0} \frac{b_i - a_i^T x_k}{a_i^T p_k}$ and if $\bar{j} = argmin \alpha_k$, **for next iteration:** $\mathcal{W}_{k+1} = \mathcal{W}_k \cup \{\bar{j}\}$
  - Else if $\alpha_k = 1$, **for next iteration:** $\mathcal{W}_{k+1} = \mathcal{W}_k$